

PATN REFERENCE LIST

GENERAL REFERENCES

- Belbin, L. (1985). Making sense of your data. *CSIRO Industrial Research Newsletter* **171**, 5-8. (A light-hearted introduction)
- Belbin, L. (1985). A background briefing in numerical taxonomy, or does your handbag match your shoes. DECUS Symposium, Hobart, Australia, 15-22. (Another light-hearted introduction)
- Clifford, H.C and Stephenson, W.C (1975). *An Introduction to Numerical Classification*. Wiley, London. (Somewhat of a classic that is still well worth reading).
- Jardine, N. & Sibson R. (1971). *Mathematical Taxonomy*. Wiley, London. 286p.
- Legendre, L. and Legendre, P. (2012). *Numerical ecology*. Elsevier Science & Technology. UK. 1006 pages. (While I don't agree with a number of their ideas, these two have contributed a lot to the domain over many years).
- Mardia, K.V., Kent, J.T. and Bibby, J.M. (1979). *Multivariate analysis*. Academic Press. 521p.
- Sneath, P.H.A. and Sokal, R.R. (1973). *Numerical Taxonomy*. (Freeman). (One of the books that started it all).
- H. M. Pereira *et al.* (2013). Essential biodiversity variables. *Science*, **339**, 277-278. (A very recent stimulating contribution to the domain).

ASSOCIATION

- Austin, M.P. and Belbin, L. (1980). The application of the TWO-STEP algorithm to two floristic data sets. *CSIRO Land Use Research Technical Memorandum* **80/20**. (two-step, part 2).
- Austin, M.P. and Belbin, L. (1982). A new approach to the species classification problem in floristic analysis. *Aust.J.Ecol.*, **7**, 75-89. (two-step, part 3).
- Belbin, L. (1980). TWO STEP : a program incorporating asymmetric comparisons that uses two steps to form a dissimilarity matrix. *CSIRO Land Use Research Technical Memorandum* **80/9**. (Two-step part 1).
- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. *CSIRO Division of Water and Land Resources Technical Memorandum* **84/23**. (Covers most of the algorithms in Pattern Analysis).
- Bray, J.R. and Curtis J.T. (1957). An ordination of the upland forest communities of southern Wisconsin, *Ecological Monographs*, **27**, 325-349. (The classic association measure).
- Czekanowski J (1913): Zarys method statystyczny. Warsaw. (Yes, old, but original and just to demonstrate that old isn't out of date because this work has been reinvented a dozen times since).
- De'ath, G. (1999). Extended dissimilarity: a method of robust estimation of ecological distances from high beta diversity data. *Plant Ecology* **144**, 191-199. (An interesting alternative to under-estimation of association. Glenn has done some good work in the domain).
- Faith, D.P. (1983). Asymmetric binary similarity measures. *Oecologia* **57**, 287-290. (Reminiscent of the asymmetry debate).
- Faith, D.P. (1984). Patterns of sensitivity of association measures in numerical taxonomy. *Mathematical Biosciences* **69**, 199-207.
- Faith, D.P., Austin, M.P., Belbin, L. and Margules, C.R. (1985). Numerical classifications of profile

- attributes in environmental studies. *J. Env. Management*, **20**, 73-85. (Will get you thinking about 2d variables).
- Faith, D.P., Minchin, P.R. and Belbin, L (1987). Compositional dissimilarity as a robust measure of ecological distance: A theoretical model and computer simulations. *Vegetatio* **69**, 57-68. (This was the paper that discovered the nature of under-estimation of association – something we didn't expect in our study).
- Gower, J.C. (1971). A general coefficient of similarity and some its properties. *Biometrics* **27**, 857-71. (John Gower is a giant in the field and this is his paper on the Gower Metric association measure).
- Jaccard, P. (1908). Nouvelles recherches sur la distribution florale. *Bull.Doc.Vaud.Sci.Nat.*, **44**, 223-270. (The classic paper on the Jaccard measure. Is it simpler than Bray and Curtis/Czekanowski? Not really).
- Kendall, D.G. (1970). Seriation from abundance matrices. In Hodson, F.R. and Kendall, D.G. (eds.). *Mathematics in the archaeological and historical sciences*. Edinburgh University Press, 215-252. (1d ordination).
- Tenenbaum, J.B., de Silva, V. & Langford, J.C. (2000). A global network framework for nonlinear dimensionality reduction. *Science* **290**, 2319-2323.

CLASSIFICATION

- Anderberg, M.R. (1973). *Cluster Analysis for Applications*. Academic Press, New York. 359p.
- Austin, M.P. and Belbin, L. (1982). An analysis of succession along an environmental gradient using data from a lawn. *Vegetatio*. **46**, 19-30. (Bit of a classic Mike Austin study. Good read)
- Belbin, L. (1984). FUSE. A FORTRAN program for agglomerative fusion. *Computers and Computing*, **10**(4), 361-284. (This was my approach to classification that PATN is still based on).
- Belbin, L. (1987). The use of non-hierarchical allocation methods for clustering large sets of data. *Aust.Computer J.*, **19**(1), 32-41. (This is still one of the most efficient clustering algorithms, and currently in PATN).
- Belbin, L. (1990). The analysis of pattern in bio-survey data. In *Nature Conservation: Cost Effective Biological Surveys and Data Analysis*, Margules, C.R. & Austin, M.P. (eds.). CSIRO, Melbourne. (A summary of Pattern Analysis toward biodiversity conservation).
- Belbin, L. (1992). Comparing two sets of community data, a method for testing reserve adequacy. *Australian.J.Ecology*, **17**, 255-262. (Complementarity and representativeness, more).
- Belbin, L. (1993). Environmental representativeness: regional partitioning and reserve selection. *Biological Conservation*, **66**, 223-230. (The key paper on representativeness and complementarity).
- Belbin, L. & Austin, M.P. (1990). ECOSIM: A simulation model for training in cost-effective survey methods. In *Nature Conservation: Cost Effective Biological Surveys and Data Analysis*, Margules, C.R. & Austin, M.P. (eds.). CSIRO, Melbourne. (How we can tell what methods work).
- Belbin, L., Marshall, C. and Faith, D.P. (1983). Representing relationships by the automatic assignment of colour. *Australian Computer J.*, **15**(4), 160-163. (Probably my favourite simple paper!)
- Belbin, L., Faith, D.P. and Milligan, G.W. (1992). A comparison of two approaches to β -flexible clustering. *Multivariate Behavioural Research*. **27**, 417-433. (Some nice work with a colleague on hierarchical classification evaluation using simulated data).
- Belbin, L. & McDonald, C. (1993). Comparing three classification methods for use in Ecology. *J.*

- Vegetation Science*, **4**, 341-348. (As it says).
- Dudman, W.F. and Belbin, L. (1987). A numerical taxonomic analysis of some Rhizobium strains using a qualitative coding of immunodiffusion reactions. *Applied & Environmental Microbiology*, **1825-1830**. (A case study).
- Everitt, B. (1980). *Cluster Analysis*. 2nd Ed. (Heinemann Educational for Social Science Research Council: London). 136p.
- Faith, D.P. and Belbin, L. (1986). Complexity, consensus, and the comparison of classifications. *J.Classification*, **3**(2), 255-277.
- Gill, A.M., Belbin, L. and Chippendale, G.M. (1985). Phytogeography of EUCALYPTUS in Australia. *Bureau of Flora and Fauna, Australian Flora and Fauna Series*, **3**. Canberra, 53p. (A nice example of classification of Australian Eucs).
- Gower, J. C. (1967): A comparison of some methods of cluster analysis. *Biometrics*, **23**(4), 623-637.
- Hartigan, J.A. (1975). *Clustering Algorithms*. (Wiley).
- MacNaughton-Smith, P. (1965). Some statistical and other numerical techniques for classifying individuals. Home Office Research Unit Report 6, (HMSO: London).
- MacNaughton-Smith, P. Williams, W. T. and Dale, M.B. (1964). Dissimilarity analysis: a new technique of hierarchical sub-division. *Nature*, **202**, 1034-1035.
- Margules C., Faith D. P. and Belbin L. (1985). An adjacency constraint in agglomerative hierarchical clustering of geographical data. *Environment and Planning A*, **17**, 397-412. (Interesting outcomes from constrained classification).
- Margules, C.R. & Belbin, L. (1990). Where should climate monitoring stations be? *Proceedings, Workshop on IGBP Network of Global Change Regional Research Centres: Scientific Problems and Concept Developments*, Warsaw, September, 1989. (An example of the flexibility of pattern analysis in solving unusual problems).
- McKenzie, N.L., Robinson, A.C., Gungkyo, A. and Belbin L. (1987). Methods. In, McKenzie, N.L., Robinson, A.C. (eds.), *a biological survey of the Nullarbor region, South and Western Australia in 1984*. South Australian Department of Environment and Planning, 32-38. (This is one of a suite of papers that explain the analysis of bio-survey data for establishing bio-reserves on the Nullarbor Plain).
- McKenzie, N.L., Belbin, L., Gungkyo, A. and Robinson, A.C. (1987). Analysis of biophysical patterns. In McKenzie, N.L., Robinson A.C. (eds.), *a biological survey of the Nullarbor region, South and Western Australia in 1984*. South Australian Department of Environment and Planning, 211-232.
- McKenzie, N.L., Belbin, L., Margules, C.R. & Keighery, G.J. (1989). Selecting representative reserves in remote areas: a case study of the Nullarbor region, Australia. *Biological Conservation* **50**, 1-11.
- Williams, W.C and Bunt, J.S. (1981). Studies in the analysis of data from Australian tidal forests ('Mangroves'). II. The use of an asymmetric monothetic divisive classificatory program. *Aust.J.Ecol*, **5**(4), 385-90. (Classic Bill Williams, but the method while interesting isn't optimal).
- Williams, B.G., Ward, J.K. and Belbin, L (1987). Numerical classification of shallow groundwater chemistry of the Murrumbidgee Irrigation Area. *Aust.J.Soil Res*, **25**, 337-345. (A case study of classification).
- Wishart, D. (1978). *Clustan User Manual*. Third edition. Edinburgh University, Program Library Unit, Report no.47. (An oldie but goodie).

ORDINATION

- Austin, M.P., Williams, O.B. and Belbin, L. (1982). Grassland dynamics under sheep grazing in an Australian Mediterranean type climate. *Vegetatio*, **46**, 210-212. (How to handle timer series using ordination).
- Belbin, L. (1991). Semi-strong hybrid Scaling, a new ordination algorithm. *J. Vegetation Science*, **2**, 491-496. (The algorithm behind PATN's SSH).
- Coxon A.P.M. (1982). The user's guide to multidimensional scaling. Heineman, London 271p. (Good text).
- Faith, D.P., Minchin, P.R. and Belbin, L. (1985). Parsimony and falsification in Ecology: towards an assumption-free approach to the studies of species' response to gradients. In *Leemans, R., Prentice, C. and Van der Maarel, E. (eds.). Theory and Models in Vegetation Science, Studies in Plant Ecology 16, Uppsala, 31-32.*
- Gower, J.C. (1987). Introduction to ordination techniques. In L. Legendre and P. Legendre (eds.), *Developments in numerical Ecology*, Springer-Verlag, Berlin.
- Guttman L (1968). A general non-metric technique for finding the smallest co-ordinate space for a configuration of points. *Psychometrika* **33**, 469-506. (Another paper behind the utility of SSH).
- Hill, M.O. (1979). DECORANA, A FORTRAN program for detrended correspondence analysis and reciprocal averaging. Department of Ecology and Systematics, Cornell University: NY, USA. (Not my favourite by a long shot, but know thy enemy).
- Kruskal J B & Wish M (1978). *Multidimensional scaling*. Sage, California, 94p. (very readable classic).
- Kruskal J B, Young F W and Seery, J.B. (1973). How to use KYST, a very flexible program to do multidimensional scaling and unfolding. Unpublished, Bell Laboratories. (KYST manual, not fabulous but historic).
- Kruskal, J B (1962). Multidimensional scaling by optimising goodness of fit to a non-metric hypothesis. *Psychometrika*, **29**(1), 1-27.
- Kruskal, J B (1964). Non-metric multidimensional scaling: a numerical method. *Psychometrika*, **29**(2), 115-129. (The original NMDS).
- Lingoes J C & Roskam E . (1973). A mathematical and empirical analysis of two multidimensional scaling algorithms. *Psychometrika*, **38**(1), 1-81. (A technical summary).
- Ortega, J. (1967). The Givens-Householder method for symmetric matrices. In '*Mathematical Methods for Digital Computers*', Ralston, A. and Wilf, H.S. eds., **2**, 94-115. (Wiley NY).
- Rummell, J. (1967). Understanding Factor Analysis. *J. Conflict Resolution*, **11**(4), 444-80. (A good introduction to Factor Analysis).
- Shepard R N (1962). The analysis of proximities: multidimensional scaling with an unknown distance function. *Psychometrika*, **27**(2), 125-140. (An important paper)
- Shifman S S, Reynolds M L & Young F W (1981). Introduction to multidimensional scaling. Theory methods and applications. Academic Press, New York, 413p. (Lots of good examples).
- Sokal, R.R. & Rohlf, F.J. (1981). *Biometry. Principals and practice of statistics in biology*. Freeman, San Francisco. 859p. (Another classic).
- Spencer R (1986). Similarity mapping. *Byte*, August, 85-92. (A very simple introduction to multidimensional scaling).
- Wilkinson, J.H. and Reinsch, C. (1971). *Handbook for automatic computation* Volume 2, Linear Algebra (F.L. Baur ed.). Springer Verlag.